Import Data using Transform Maps (Spreadsheet)

Phase-2 -- Requirement Analysis Phase

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| Date | 29.09.2026 |
| Team Id | SWTID-2026-2101 |
| Project Name | Import Data using Transform Maps (Spreadsheet) |
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1. Requirement Analysis Overview

Primary objective: ensure accurate, consistent, and controlled import of employee records from a spreadsheet into ServiceNow.

- Expected data: a structured spreadsheet containing Employee ID, Name, Email, Department, Location.
- Import process: use Import Sets → staging table → Transform Map → target Employee Test table.
- Validation: verify field mapping, successful transformations, and transformation history.
- Duplicate handling: identify by Employee ID using Coalesce; update existing records, insert new ones, avoid duplicates on repeated imports.
- Reporting & dashboards: support standard reports (by department, by location, full list) and an Employee Analytics dashboard.
- Governance: controlled sharing and role-based access for imports and dashboard views.

## 2. Functional Requirements

- Import employee data from an Excel spreadsheet (XLSX/CSV export acceptable).
- Load spreadsheet rows into an Import Set staging table for review.
- Create a target **Employee Test** table to receive transformed records.
- Map each source spreadsheet field to the corresponding Employee Test field.
- Use a Transform Map to convert staging rows into Employee Test records.
- Validate transformed records prior to publishing to production tables.
- Insert new employee records when Employee ID is not present.
- Update existing employee records when Employee ID matches.
- Identify existing employees primarily by Employee ID as the unique identifier.
- Prevent unnecessary duplicate records through Coalesce and appropriate field mapping.
- Provide a review mechanism for import results and transformation logs.
- Support repeated employee-data imports using the existing Import Set table.
- Allow new employee data to be added to the existing import process.
- Detect changes in existing employee information such as name and email.
- Distinguish between newly inserted and updated employee records.
- Identify records that are ignored when the same data is imported again.
- Provide successful completion status for the import and transformation process.
- Allow users to review the final employee records in the Employee Test table.
- Provide employee reports based on department and location.
- Provide a detailed employee list containing Employee ID, Employee Name, Email, Department, and Location.
- Provide an Employee Analytics Dashboard containing the created reports.
- Allow dashboard information to be shared with appropriate users, groups, or roles.
- Support centralized viewing of employee data and import-related information.

## 3. Data Requirements

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- Required columns: **Employee ID, Employee Name, Email, Department, Location**.
- Employee ID should be treated as the primary record-identification field.
- Employee ID values should be available for each employee record.
- Employee IDs should follow a consistent format.
- Employee Name should be stored consistently across employee records.
- Email information should follow a consistent format.
- Department values should use standardized names.
- Location values should use standardized names.
- Column headers should be consistent and documented.
- Source columns should have corresponding fields in the Employee Test table.
- Data types should be compatible with the target fields.
- Empty or incomplete values should be identified during validation.
- Invalid or malformed data should be identified before or during transformation.
- Duplicate Employee IDs should be reviewed before importing data.
- Changes to existing employee information should be identifiable using Employee ID.
- New Employee IDs should be distinguishable from existing Employee IDs.
- Imported data should remain traceable from the source spreadsheet to the target employee record.
- Data should be reviewed after transformation to confirm successful import.
- Import results should allow inserted, updated, and ignored records to be identified.
- The final Employee Test records should contain the required employee information for reporting and dashboard purposes

4. Import and Transformation

Requirements

Conceptual flow and connectivity for the import pipeline.

- Spreadsheet data → Import Set staging table → Transform Map → Employee Test table.
- Import Set should act as a temporary staging area for incoming employee data.
- Use the existing **Employee Import** table for subsequent spreadsheet imports.
- Select the appropriate spreadsheet file as the import source.
- Maintain the correct sheet number and header row when loading spreadsheet data.
- Define clear source-to-target field mappings.
- Ensure all required source fields have corresponding target fields.
- Create a Transform Map connecting **Employee Import** to **Employee Test**.
- Use automatic field mapping where matching fields are available.
- Use Mapping Assist when manual source-to-target mapping is required.
- Ensure Employee ID is correctly mapped between the source and target tables.
- Execute the Transform Map after the import data has been loaded successfully.
- Create new Employee Test records for new Employee IDs.
- Update existing Employee Test records when matching Employee IDs are identified.
- Use Coalesce to support correct record identification during repeated imports.
- Prevent unnecessary duplicate records during repeated transformations.
- Review the transformation result after execution.
- Verify that transformed records appear correctly in the Employee Test table.
- Review transformation history to identify inserted, updated, and ignored records.
- Verify that repeated identical spreadsheet imports do not create unnecessary records.
- Maintain transformation results for review, troubleshooting, and data-quality checking.
- Confirm that the import process reaches a successful or completed state before considering the operation finished.
- Ensure the transformed employee records are available for subsequent reporting and dashboard requirements.

5. Duplicate Management

Requirements

- Use **Employee ID** as the primary means to identify existing employee records.
- Configure **Coalesce** on Employee ID to support reliable record matching.
- Employee ID should remain consistent across repeated spreadsheet imports.
- When Employee ID matches an existing record, update the existing employee record.
- When Employee ID does not exist, insert a new employee record.
- Repeated identical imports should not create additional employee records.
- Prevent duplicate records caused by repeated spreadsheet uploads.
- Detect changes in existing employee information using the Employee ID.
- Allow updated employee details such as **Employee Name** and **Email** to replace older values when the Employee ID matches.
- Ensure new Employee IDs are treated as new employee records.
- Review duplicate or conflicting Employee IDs before transformation where required.
- Ensure the field used for Coalesce is mapped correctly between the source and target tables.
- Maintain consistent Employee ID values between the spreadsheet and Employee Test table.
- Review transformation history after each import to verify duplicate handling.
- Categorize import results as **Inserted, Updated, Ignored, or Error** where applicable.
- Verify that ignored records do not create additional employee records.
- Verify that updated records retain the same Employee ID.
- Verify that newly inserted records contain the required employee information.
- Use transformation results to support auditing and troubleshooting.
- Maintain a clean and consistent Employee Test table after repeated imports.
- Validate duplicate-handling behavior by importing the same spreadsheet again and reviewing the transformation results. In the documented example, the repeated four-row import resulted in **0 inserted, 0 updated, and 4 ignored** records.

## 6. Validation Requirements

- Confirm imported records appear correctly in the **Employee Test** table after transformation.
- Verify each source field is mapped to the correct target field.
- Verify that **Employee ID** values are imported correctly.
- Verify Employee Name, Email, Department, and Location values after transformation.
- Check that the number of transformed records matches the expected import results.
- Validate that transformation rules are applied successfully.
- Review transformation history after every import.
- Check the import state to confirm successful completion.
- Review transformation results for **Inserted, Updated, and Ignored** records.
- Verify that new Employee IDs create new records.
- Verify that matching Employee IDs update existing records.
- Confirm repeated imports do not create unnecessary duplicate records.
- Verify that Coalesce identifies existing records correctly.
- Check that changed employee information is reflected in the existing record.
- Confirm that identical repeated data is handled without creating additional records.
- Identify any failed, skipped, or problematic records for further review.
- Verify that the final employee records contain the required fields.
- Confirm the imported data is suitable for generating employee reports.
- Verify employee records can be grouped by **Department** and **Location** for reporting.
- Review the final data for consistency before using it in reports and dashboards.
- Use transformation history to verify the outcome of repeated imports and duplicate handling.

7. Reporting Requirements

- Create an **Employees by Department** report to show employee distribution across departments.
- Create an **Employees by Location** report to show employee distribution across locations.
- Create an **Employee List Report** containing detailed employee information.
- Include **Employee ID, Employee Name, Email, Department, and Location** in the employee list report.
- Use appropriate aggregation such as **Count** for department and location reports.
- Use a **Pie Chart** to visualize employees by department.
- Use a **Bar Chart** to visualize employees by location.
- Provide a clear summary of the total employee records available in the Employee Test table.
- Ensure reports use the transformed employee data from the Employee Test table.
- Reports should reflect newly inserted employee records.
- Reports should reflect updated employee records.
- Reports should provide meaningful visibility into employee distribution.
- Maintain consistent report names for easy identification.
- Reports should be saved after successful configuration.
- Reports should be available for addition to the Employee Analytics Dashboard.
- Reports should support centralized monitoring of employee information.
- Report data should be reviewed to ensure that it matches the Employee Test table.
- Reports should help administrators understand employee distribution by department and location.
- The reporting structure should support further analysis of imported employee data.
- Reports should provide a simple and understandable presentation for users.
- The three reports should collectively provide **summary, distribution, and detailed employee information**.

8. Dashboard Requirements

- Create an **Employee Analytics Dashboard** as a centralized view of employee information.
- Add the **Employees by Department** report to the dashboard.
- Add the **Employees by Location** report to the dashboard.
- Add the **Employee List Report** to the dashboard.
- Present employee information in a clear and organized layout.
- Provide a quick overview of employee distribution by department.
- Provide a quick overview of employee distribution by location.
- Provide access to detailed employee information through the Employee List Report.
- Ensure dashboard reports are based on the latest available Employee Test data.
- Allow administrators to monitor employee information from a single centralized location.
- Provide visual representation of employee data through charts and reports.
- Keep the dashboard layout simple and easy for users to understand.
- Allow relevant reports to be added or removed from the dashboard when required.
- Support dashboard sharing with appropriate users, groups, or roles.
- Restrict dashboard access according to defined user or role requirements.
- Ensure dashboard users only receive access appropriate to their responsibilities.
- Provide a centralized view that reduces the need to check individual reports separately.
- Use the dashboard to support employee-data monitoring and analysis.
- Review dashboard information after imports to ensure that the displayed data reflects the Employee Test table.
- Support future expansion with additional employee-related reports if required.
- Ensure the dashboard brings together **department, location, and detailed employee information** in one view.

9. Learning & 10. Requirement Summary

Learning requirements

- Understand the purpose of **Import Sets** in ServiceNow.
- Learn how Import Sets act as a staging area for external data.
- Understand how spreadsheet data is loaded into an Import Set.
- Understand the purpose of **Transform Maps**.
- Learn how source fields are mapped to target fields.
- Understand the relationship between the **Employee Import** and **Employee Test** tables.
- Learn how **Auto Map Matching Fields** can assist with field mapping.
- Understand the purpose of **Mapping Assist**.
- Understand the role of **Coalesce** in identifying existing records.
- Learn how Employee ID can be used to identify employee records.
- Understand the difference between **new records and existing records**.
- Learn how new employee records are inserted.
- Learn how existing employee records are updated.
- Understand how repeated imports are handled.
- Learn how duplicate records can be prevented during repeated imports.
- Understand **Inserted, Updated, and Ignored** import results.
- Learn how to review transformation results after an import.
- Practice checking imported records for correctness.
- Learn how to identify incorrect or incomplete employee information.
- Understand the importance of consistent employee data.
- Learn how employee data can be grouped by **Department** and **Location**.
- Understand the purpose of basic ServiceNow reports.
- Learn how to create reports using transformed employee data.
- Understand how multiple reports can be combined into a dashboard.
- Learn how dashboards provide a centralized view of information.
- Understand how dashboard access can be controlled for users, groups, and roles.
- Develop basic troubleshooting skills for import and transformation issues.
- Gain practical understanding of the overall employee data import process.

Requirement Summary

Overall Requirement Flow

**Spreadsheet → Import Set → Transform Map → Coalesce → Employee Records → Validation →**

**Reports → Dashboard**

Summary Points

- Import employee information from a structured spreadsheet.
- Use an **Import Set** to stage the imported employee data.
- Use a **Transform Map** to define source-to-target field relationships.
- Transform the staged data into the **Employee Test** table.
- Use **Employee ID** as the main identifier for employee records.
- Apply **Coalesce** to identify matching employee records.
- Insert records when the Employee ID is new.
- Update records when the Employee ID already exists.
- Prevent unnecessary duplicate records during repeated imports.
- Identify inserted, updated, and ignored records after transformation.
- Validate employee information after the transformation process.
- Ensure Employee ID, Employee Name, Email, Department, and Location are available in the target records.
- Use the transformed employee data for reporting.
- Create department-wise employee analysis.
- Create location-wise employee analysis.

- Provide a detailed employee list report.
- Combine the reports into an **Employee Analytics Dashboard**.
- Provide a centralized view of employee information.
- Control dashboard access for appropriate users, groups, or roles.
- Support review and monitoring of imported employee information.
- Provide a repeatable process for future employee data imports.
- Maintain consistent employee information between repeated imports.
- Support basic analysis of employee distribution by department and location.
- Provide practical learning of ServiceNow data import and transformation concepts.
- Establish a clear foundation for implementing the project in later phases.